

Tide retrospective: the expansion pairing reads off the quartic coefficient

Tide loop (Claude Fable 5, with GPT-5.6 Sol deliberation)

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1 Setting

The germbij note's Section 7.4 recovers the subleading coefficients of an analytic potential from the corrections of the Laplace expansion at their distinct orders. The recovery thread had so far certified the leading data (degree and scale, exactly and asymptotically) and a monotonicity shortcut for signed perturbations; the expansion *mechanism* itself, the pairing the note actually describes, had no Lean instance. This tide provides the smallest one: the first correction of the quartic-perturbed Gaussian, with an explicit error bound, and the recovery of the quartic coefficient from it.

2 The thing formalised

Three declarations in `Laplace/OneD/RecoveryExpansion.lean`, sorry-free with zero warnings. The elementary remainder bound¹: for $s \geq 0$, $0 \leq e^{-s} - 1 + s \leq s^2$, proved with two applications of $1 + x \leq e^x$ and no calculus. The expansion²: for $b \geq 0$, $t > 0$,

$$0 \leq Z_b(t) - \sqrt{2\pi} t^{-1/2} + 3b\sqrt{2\pi} t^{-3/2} \leq 105 b^2 \sqrt{2\pi} t^{-5/2}, \quad Z_b(t) = \int e^{-t(x^2/2+bx^4)} dx.$$

And the recovery³: if Z_{b_1} and Z_{b_2} eventually agree ($b_i \geq 0$), then $b_1 = b_2$. The numerical check (executed, archived in the tide log) confirms both the limit $-3b\sqrt{2\pi}$ of the rescaled correction and the error constant with the expected factor-two slack from the elementary bound.

3 Proof strategy

Split $e^{-t(x^2/2+bx^4)} = q \cdot e^{-tbx^4}$ with q the Gaussian factor, and expand $e^{-tbx^4} = 1 - tbx^4 + r$ with $0 \leq r \leq (tbx^4)^2$ pointwise. The seabed's scale- t moments (the $k = 0, 2, 4$ cases of $\int x^{2k} q = (2k-1)!! \sqrt{2\pi} t^{-(k+1/2)}$, so 1, 3, 105) do the rest: the $k = 0$ and $k = 2$ terms cancel the main and correction terms of the expansion *exactly*, so the error is literally $\int q r$, bounded two-sidedly by the $k = 4$ moment. No substitution, no dominated-convergence, no asymptotic machinery. For the recovery, the difference of the two expansion errors is exactly $3(b_1 - b_2)\sqrt{2\pi} t^{-3/2}$ when the partition functions agree, and the two error bounds squeeze it below order $t^{-5/2}$; one archimedean choice of t closes the contradiction, with no limits taken.

¹`exp_neg_sub_one_add_bounds.`

²`quartic_partition_expansion_bounds.`

³`quartic_coefficient_recovery_of_eventuallyEq.`

4 Roadblocks and resolutions

Four build iterations, all elaboration-level, all of documented classes: a missing import for the seabed integrability family; the `Pi.add` gotcha in `integral.add` (fixed by the CLAUDE.md’s type-ascribed single-lambda witnesses); brittle `rw`-into-closed-form steps replaced by the robust pattern “`norm_num [Nat.doubleFactorial]` at hypothesis and goal, then `linarith` with the integral as an atom”; and a `calc` of equalities against a $\leq$ goal needing `le_of_eq`.

5 What was Mathlib, what was new

Mathlib: $1 + x \leq e^x$, integral linearity, `integral_mono`, domination. Seabed: the scale- t Gaussian moments and the monomial integrability family, both from the May tides, doing exactly the work they were built for. New: the observation that the moment cancellation is exact (the expansion error *is* the remainder integral), which makes the whole development bookkeeping-free.

6 Lessons

When a closed form is consumed, normalize both sides to a common normal form and finish with `linarith` treating opaque terms as atoms; chasing exact `rw` orientations against `norm_num` output is the fragile alternative. The elementary s^2 remainder bound (constant twice the sharp $s^2/2$) is the right trade: it removes all calculus from the file at the cost of a factor the application never notices.

7 Follow-ups

- The general inductive step (all subleading coefficients of an analytic perturbation, correction by correction) would need expansions to arbitrary order: the natural shape is a finite Taylor expansion of e^{-s} with the same elementary remainder trick, and it would subsume this tide’s $k = 1$ instance.
- Tag the note’s Section 7.4 dimension-one item with the expansion recovery once merged; it certifies the pairing mechanism specifically.